

Tarea Sección 6.6

3) a) $\sin^{-1} \frac{\sqrt{3}}{2} = \frac{\pi}{3}$

b) $\cos^{-1} \frac{\sqrt{2}}{2} = \frac{\pi}{4}$

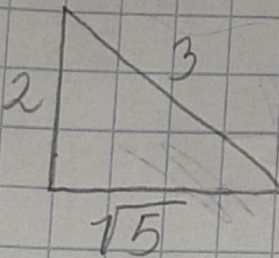
c) $\tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$

7) a) $\sin[\sin^{-1}(-\frac{3}{10})] = -\frac{3}{10}$

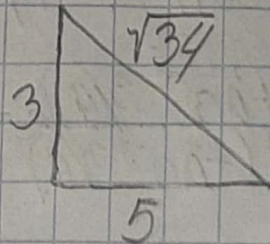
b) $\cos[\cos^{-1}(1/2)] = \frac{1}{2}$

c) $\tan(\tan^{-1} 14) = 14$

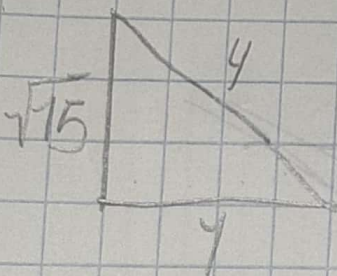
5) a) $\cot(\sin^{-1} 2/3) = \frac{\sqrt{5}}{2}$



b) $\sec[\tan^{-1}(-\frac{3}{5})] = \frac{\sqrt{34}}{5}$

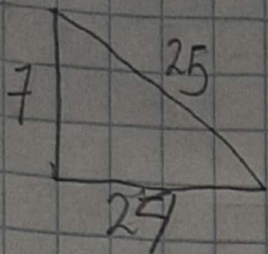


c) $\csc[\cos^{-1}(-\frac{1}{4})] = \frac{4}{\sqrt{15}}$



$$19) a) \sin[2\cos^{-1}(-\frac{24}{25})] =$$

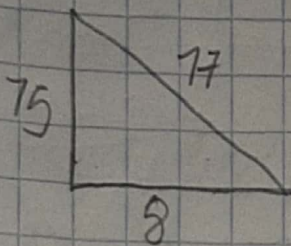
$$2\sin(\cos^{-1}(-\frac{24}{25}))\cos(\cos^{-1}(-\frac{24}{25}))$$



$$2(\frac{7}{25})(\frac{24}{25}) = -\frac{336}{625}$$

$$b) \cos(2\sin^{-1}\frac{15}{17})$$

$$\cos^2(\sin^{-1}\frac{15}{17}) - \sin^2(\sin^{-1}\frac{15}{17})$$



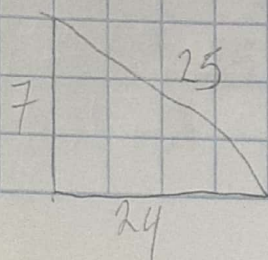
$$(\frac{8}{17})^2 - (\frac{15}{17})^2 = -\frac{167}{289}$$

$$c) \tan(2\tan^{-1}\frac{3}{4})$$

$$\frac{2\tan(\tan^{-1}\frac{3}{4})}{1 - \tan^2(\tan^{-1}\frac{3}{4})} = \frac{2(\frac{3}{4})}{1 - (\frac{3}{4})^2} = \frac{24}{7}$$

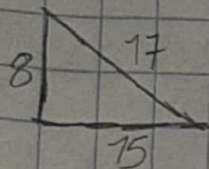
$$21) \sin[\frac{1}{2}\sin^{-1}(-\frac{7}{25})]$$

$$\frac{1}{2} \sqrt{1 - \cos(\sin^{-1}(-\frac{7}{25}))} = \frac{1}{2} \sqrt{1 - \frac{24}{25}} = -\frac{\sqrt{2}}{10}$$



$$b) \cos\left(\frac{1}{2} \tan^{-1} \frac{8}{15}\right)$$

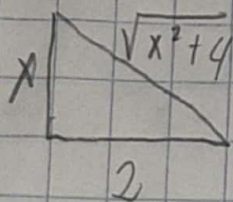
$$\pm \frac{1 + \cos(\tan^{-1} 8/15)}{2} = \frac{1 + \frac{15}{17}}{2} = \frac{4 + \sqrt{17}}{17}$$



$$c) \tan\left(\frac{1}{2} \cos^{-1} \frac{3}{5}\right)$$

$$\pm \frac{1 - \cos(\cos^{-1} 3/5)}{\sqrt{1 + \cos(\cos^{-1} 3/5)}} = \pm \frac{1 - 3/5}{\sqrt{1 + 3/5}} = \frac{1}{2}$$

$$27) \sec\left(\sin^{-1} \frac{x}{\sqrt{x^2+4}}\right) =$$

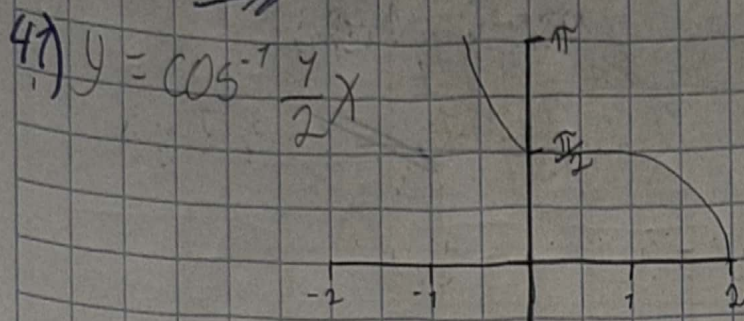


$$= \frac{\sqrt{x^2+4}}{2}$$

$$= 2$$

$$= \frac{\sqrt{x^2+4}}{2}$$

35) a) $-\frac{\pi}{2}$ b) 0 c) $\frac{\pi}{2}$



51) $y = \frac{7}{2} \sin^{-1}(x-3)$

a) $\mathbb{D} = [2, 4]$

b) $\mathbb{R} = -\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$

c) $x = \sin 2y + 3$

59) $\frac{\sin x}{3} = \frac{\sin y}{4}$

$x = \pi - x$

$x = \sin^{-1}\left(\frac{3}{4} \sin y\right)$

67) $6 \sin^3 \theta + 18 \sin^2 \theta - 5 \sin \theta - 15 = 0; \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$6 \sin^2 \theta (\sin \theta + 3) - 5(\sin \theta + 3) = 0$

$\sin \theta + 3 (6 \sin^2 \theta - 5) = 0$

$$57) y = 15 - 2 \cos x; [0, \pi]$$

$$x = \cos^{-1} \left[\frac{1}{2} (15 - y) \right]$$

$$\begin{aligned}\sin \theta + 3 &= 0 \\ \theta &= \sin^{-1} -3 \\ \theta &= \text{no existe}\end{aligned}$$

$$\begin{aligned}6 \sin^2 \theta - 5 &= 0 \\ \sin^2 \theta &= \frac{5}{6} \\ \sin \theta &= \pm \sqrt{\frac{5}{6}}\end{aligned}$$

$$\sin \theta = \pm \sqrt{\frac{30}{6}} = \pm 1, \text{ B03}$$

$$71) 3 \cos 2x - 7 \cos x + 5 = 0 \quad ; \quad [0, 2\pi)$$

$$\begin{aligned}3(\cos^2 x - \sin^2 x) - 7 \cos x + 5 &= 0 \\ 3(\cos^2 x - (1 - \cos^2 x)) - 7 \cos x + 5 &= 0 \\ 3(2 \cos^2 x - 1) - 7 \cos x + 5 &= 0 \\ 6 \cos^2 x - 3 - 7 \cos x + 5 &= 0 \\ 6 \cos^2 x + 2 - 7 \cos x &= 0\end{aligned}$$

$$\begin{aligned}2 \cos x - 1 &= 0 \\ \cos x &= \frac{1}{2} \\ x &= \cos^{-1} \frac{1}{2} = \frac{\pi}{3}, \frac{5\pi}{3}\end{aligned}$$

$$(2 \cos x - 1)(3 \cos x - 2) = 0$$

$$\begin{aligned}3 \cos x - 2 &= 0 \\ \cos x &= \frac{2}{3} \\ x &= \cos^{-1} \frac{2}{3} = 48^\circ\end{aligned}$$

$$\frac{\pi}{3} = 1.047$$

$$\frac{5\pi}{3} = 5.236$$

$$83) \tan^{-1} x + \tan^{-1} \frac{1}{x} = \frac{\pi}{2}$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{x + \frac{1}{x}}{1 - x(\frac{1}{x})} = \frac{x + \frac{1}{x}}{0}$$

$$x + \frac{1}{x} = \alpha + \beta = \frac{\pi}{2}$$